

Q3. Given, Y and X are random variables
conditional expectation of Y given X is linear
 $E(Y|X) = \alpha + \beta X$

①

$$\text{evid, } \varepsilon = Y - E(Y|X) \\ = Y - \alpha - \beta X \quad (\text{from question})$$

$$\varepsilon + \alpha + \beta X = Y$$

$$\therefore Y = \alpha + \beta X + \varepsilon \quad (\text{using evid term})$$

as ε satisfies $E(\varepsilon|X) = 0$

$$\text{proof, } E(\varepsilon|X) = E(Y - E(Y|X)|X) \\ = E(Y|X) - E(Y|X) \\ = 0$$

② from ①, $Y = \alpha + \beta X + \varepsilon$

$$\text{Cov}(Y, X) = \text{Cov}(\alpha + \beta X + \varepsilon, X)$$

typically $\text{Cov}(\varepsilon, X)$ is assumed zero in regression models

$$\text{Cov}(\beta\alpha, X) = 0 \quad \text{as } \alpha \text{ is constant parameter}$$

$$\therefore \text{Cov}(Y, X) = \text{Cov}(\beta X, X) \\ = \beta \text{Cov}(X, X)$$

$$\text{As } \text{Cov}(X, X) = \text{Var}(X)$$

$$\text{Cov}(Y, X) = \beta \text{Var}(X)$$

$$\text{for parameter, } \beta = \frac{\text{Cov}(Y, X)}{\text{Var}(X)}$$

Now for α , we can take expectation as
 $E(\alpha) = \alpha$ to get the parameter α value

$$Y = \alpha + \beta X + \epsilon \quad (\text{from part 1})$$

$$E(Y) = E(\alpha + \beta X + \epsilon)$$

$$= \alpha + E(\beta X) + 0 \quad (\because E(\epsilon) = 0)$$

$$= \alpha + \beta E(X)$$

$$= \alpha + \frac{E(X) \text{Cov}(Y, X)}{\text{Var}(X)} \quad (\text{from part 2})$$

$$\therefore \alpha = E(Y) - \frac{E(X) \text{Cov}(Y, X)}{\text{Var}(X)}$$

③ In this case, linear model doesn't necessarily hold.

To find minimum value of $f(x)$,

$$f'(x) = 0$$

$$\frac{\partial}{\partial \alpha} [E\{(Y - \alpha - \beta X)^2\}] = 0 \quad (\text{we take w.r.t } \alpha \text{ as we want to find } \alpha \text{ for min value})$$

$$\frac{\partial}{\partial \alpha} (E(Y^2) + E(\alpha + \beta X)^2 - 2E(\alpha - \beta X)) = 0$$

$$\frac{\partial}{\partial \alpha} (E(\alpha^2 + \beta^2 X^2 + 2\alpha\beta X) - 2E(\alpha) - 2E(\beta X)) = 0$$

$$\frac{\partial}{\partial \alpha} (E(2\alpha\beta X) - 2E(\alpha)) = 0$$

$$\frac{\partial}{\partial \alpha} (E(Y^2) - 2\alpha E(Y) - 2\beta E(YX) + \alpha^2 + 2\alpha \beta E(X) + \beta^2 E(X^2)) = 0$$

$$\frac{\partial}{\partial \alpha} (-2E(Y) + \alpha^2 + \beta^2 E(X^2)) = 0$$

$$-2E(Y) + 2\alpha + 2\beta E(X) = 0$$

$$\alpha = E(Y) - \beta E(X)$$

$$\text{for } \beta, \quad \frac{\partial}{\partial \beta} [E(Y - \alpha - \beta X)^2] = 0$$

$$\frac{\partial}{\partial \beta} [E(Y^2) - 2\alpha E(Y) - 2\beta E(YX) + \alpha^2 + 2\alpha \beta E(X) + \beta^2 E(X^2)] = 0$$

$$\frac{\partial}{\partial \beta} (-2\beta E(YX) + 2\alpha \beta E(X) + \beta^2 E(X^2)) = 0$$

$$-2E(YX) + 2\alpha E(X) + 2\beta E(X^2) = 0$$

$$\beta = \frac{E(YX) - \alpha E(X)}{E(X^2)}$$

$$\text{Substituting } \alpha = E(Y) - \beta E(X)$$

$$\beta = \frac{E(YX) - (E(Y) - \beta E(X))E(X)}{E(X^2)}$$

Rearranging we get, $\beta = \frac{E(YX) - E(Y)E(X)}{[E(X)^2] - [E(X)]^2}$

$$= \frac{\text{COV}(X, Y)}{\text{Var}(X)}$$

Similarly, $\alpha = E(Y) - \beta E(X)$
 $= E(Y) - \frac{\text{COV}(X, Y) E(X)}{\text{Var}(X)}$